

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I M. Pranay (192525382) _certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay_____

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

ANSWERS

1. Retail Data Optimization Problem

A big data architecture can integrate sales, inventory, and customer behavior data from physical stores and the online platform.

Architecture:

Data Sources → Data Ingestion → Data Storage → Big Data Processing → Analytics/ML → Dashboard & Recommendations

- **Data Sources:** POS systems, website, mobile app, inventory systems, CRM and customer activity.
- **Data Ingestion:** Apache Kafka can collect real-time sales and inventory events.
- **Storage:** Hadoop HDFS or a cloud data lake can store structured and unstructured data.
- **Processing:** Apache Spark can process large-scale batch and streaming data.
- **Real-Time Inventory:** Stream processing updates inventory whenever a product is sold or received.
- **Demand Forecasting:** Machine learning models such as regression, time-series forecasting or Random Forest can predict future product demand.
- **Personalized Marketing:** Collaborative filtering and customer behavior analysis can recommend products.
- **Visualization:** Dashboards can display inventory levels, sales trends and predicted demand.
- **Security:** Authentication, encryption and role-based access control should protect customer and business data.

Result: The architecture provides real-time inventory visibility, improves demand forecasting and enables personalized marketing.

2. Social Media Fraud Detection Problem

The platform needs a scalable system capable of processing billions of posts, comments, likes and shares.

Architecture:

User Interactions → Kafka → Stream Processing → Feature Extraction → ML Models → Fraud/Misinformation Detection → Alerts

- **Apache Kafka** collects billions of real-time user events.
- **Apache Spark Streaming or Apache Flink** processes the incoming data.
- User features such as account age, posting frequency, interaction patterns and network behavior are extracted.
- Machine learning models identify suspicious accounts.
- Classification algorithms can detect potentially misleading or harmful content.
- Natural Language Processing can analyze text in posts and comments.
- Graph-based analytics can identify groups of fake or coordinated accounts.

- Detected suspicious activities are sent to moderators or automated response systems.
- Distributed processing allows the system to scale across multiple servers.

Role of Machine Learning:

- Detects unusual user behavior.
- Classifies suspicious accounts.
- Identifies patterns associated with misinformation.
- Continuously improves using historical labeled data.

Result: Combining stream processing and machine learning enables **fast, scalable and near-real-time fraud and misinformation detection.**

3. Government Data Platform Problem

A national government data platform must integrate census, economic and geospatial information while maintaining interoperability, privacy and security.

Proposed Strategy

Data Sources → Data Integration → Data Lake → Processing → Data Governance → Secure Data Sharing

- Collect data from different government departments.
- Use standardized data formats and common data definitions.
- Apply **ETL/ELT** processes to integrate data.
- Store large datasets in a centralized data lake.
- Use metadata management to describe datasets.
- Maintain a common data catalog so departments can discover available data.
- Use APIs to enable interoperability between government systems.
- Apply data validation and quality checks.
- Use encryption for data at rest and in transit.
- Implement role-based access control.
- Apply authentication and authorization.
- Anonymize or pseudonymize sensitive citizen information.
- Maintain audit logs to track data access.
- Define data ownership, retention and sharing policies.

Result: A strong governance framework provides **interoperability, reliable data sharing, privacy protection and secure access** across government agencies.

4. Banking Fraud Detection Problem

A bank requires a low-latency pipeline capable of analyzing millions of transactions and identifying fraud in real time.

Big Data Pipeline

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Score → Alert/Block

- Transaction data is continuously collected from banking systems.
- **Apache Kafka** acts as the real-time data ingestion layer.
- **Apache Flink or Spark Streaming** processes transactions with low latency.
- Historical transaction information is stored in a data lake or distributed database.
- Features such as transaction amount, location, time, device, account history and transaction frequency are generated.
- Machine learning algorithms analyze transaction behavior.
- **Random Forest, Gradient Boosting or Logistic Regression** can be used for classification.
- Anomaly detection can identify unusual transaction patterns.
- Each transaction receives a fraud probability or risk score.
- High-risk transactions can be blocked or sent for additional verification.
- Historical data is used to retrain and improve the models.

Security

- Encrypt financial data.
- Use strong authentication and authorization.
- Maintain audit logs.
- Protect customer information.
- Monitor the system continuously.

Result: The pipeline provides **real-time, low-latency and accurate fraud detection** while supporting large transaction volumes.

5. Agriculture Smart Farming Problem

Precision agriculture can combine sensor, weather, drone and crop data to provide farmers with useful real-time insights.

Proposed Architecture

Sensors/Drones/Weather Data → Data Ingestion → Distributed Storage → Processing → Analytics/ML → Farmer Dashboard

- **Soil sensors** collect moisture, temperature, pH and nutrient information.
- **Weather stations** provide rainfall, temperature and humidity data.
- **Drones** capture crop images and field conditions.
- IoT gateways collect sensor information from different locations.
- Kafka or an IoT data platform can collect real-time data.
- A cloud data lake can store sensor, image and historical crop data.
- Apache Spark can process large datasets.
- Machine learning can analyze crop conditions and predict yield.
- Image analysis can identify crop diseases or stressed plants.
- Soil moisture analytics can determine irrigation requirements.
- Weather predictions can support decisions about irrigation and pesticide application.
- Historical crop data can be used for yield prediction.

Decision-Making Support

- **Irrigation:** Apply water based on soil moisture.
- **Fertilization:** Identify areas requiring nutrients.
- **Disease Detection:** Detect unhealthy crops using drone images.
- **Yield Prediction:** Estimate expected crop production.
- **Resource Optimization:** Reduce unnecessary water, fertilizer and pesticide usage.
- **Weather Planning:** Adjust farming activities according to weather conditions.